Before the First Token: Auditing Affective Modality Conflict in Pre-generation States

***Abstract*—Multimodal large language models deployed in affect-sensitive dialogue systems often commit to a single affective interpretation even when visual and audio/text cues imply incompatible emotional states. This paper proposes a pre-generation measurement framework that treats affective modality conflict as a structural event operationalized at a fixed interface: the hidden state at the last conditioning token before content generation begins.**

The framework yields three auditable signals. A template-sensitivity gate filters measurement instability. A split-strength metric quantifies separation between modality-conditional representations. An arbitration-bias metric characterizes the proximity of the fused representation to either modality, interpreted only on the split-valid subset. Thresholds are calibrated per backbone on a consistency domain and frozen for evaluation, yielding a four-state taxonomy: Confusion, Consensus, Balanced, and Dominant.

Experiments on multimodal sentiment datasets with graded cross-modal inconsistency support two findings. First, split strength exhibits a monotonic dose–response with conflict level. Second, within the split-valid subset, arbitration bias exposes backbone-specific signatures not predictable from conflict dosage alone. The resulting state labels provide an auditable interface for downstream policies that condition response behavior on what is structurally supported by the pre-generation state.

***Index Terms*—Multimodal sentiment analysis, Affective conflict detection, Multimodal large language models, Internal-state interpretability.**

I. Introduction

MULTIMODAL large language models are increasingly deployed as the generative backbone of dialogue systems sensitive to affect, including applications in mental health support, educational tutoring, and customer service [1], [2]. A recurring challenge in these settings is *affective modality conflict*: visual and audio/text cues of a user may convey mutually inconsistent emotional signals within a single turn. Consider a user who says "I'm fine" with a calm tone but whose facial expression suggests distress. A system that collapses such contradictory evidence into a single, unqualified affective interpretation risks inappropriate responses—not because it fails to generate fluent text, but because it commits to an affective claim without accounting for disagreement in the underlying evidence. In mental health applications, such blind commitment may inadvertently invalidate user experiences or miss signals of distress masked by social presentation [3].

A structural constraint of autoregressive generation exacerbates this problem. Once the first content token is produced, subsequent generation follows a single trajectory causally downstream of the model's initial commitment. Post-generation rationales may appear coherent yet remain agnostic to whether the pre-generation state reflected a stable consensus, a structural split between modalities, or mere instability under minor prompt variation [4].

Existing multimodal affective computing approaches address cross-modal disagreement by optimizing fusion architectures. Methods based on modality alignment, adaptive weighting, and triage networks improve prediction when sentiment polarities differ across modalities [5], [6], [27]. These contributions optimize how a model fuses evidence to produce better outputs; we ask a different question: whether the model's internal state at the moment of commitment structurally supports a single-valued affective assertion.

We operationalize affective modality conflict at a pre-generation interface, denoted t_0: the hidden state at the last conditioning token before the model commits to its first generated token. At t_0, the model has processed the full multimodal conditioning sequence but has not yet collapsed it into a single output trajectory. We summarize this pre-generation state by an auditable triplet $(S, \mathcal{D}, \mathcal{R})$: S captures template sensitivity, $\mathcal{D}$ measures cross-modal separation, and $\mathcal{R}$ reports the arbitration position of the joint summary relative to the unimodal endpoints. Together they induce a four-state taxonomy (*Confusion*, *Consensus*, *Balanced*, *Dominant*) that distinguishes stable agreement from structural disagreement and reveals when fusion exhibits directional modality preference. These state labels provide an auditable interface for conditioning downstream response behavior, enabling abstention under instability and conflict-aware interaction when modalities disagree, as illustrated in Fig. 1.

Our contributions are threefold:

- We introduce a pre-generation audit interface at t_0 for probing multimodal integration states causally upstream of any generated content.
- We formalize an auditable triplet $(S, \mathcal{D}, \mathcal{R})$ that induces a four-state taxonomy to characterize stability, split, and arbitration bias across backbones and the two bimodal protocols (V, A) and (V, T).
- We provide empirical evidence on graded cross-modal inconsistency showing a monotonic split dose–response and backbone-specific arbitration signatures, and demonstrate how the resulting state labels can condition response policies.

The remainder of this paper is organized as follows. Section II reviews related work. Section III formalizes the measurement protocol. Section IV presents experimental results. Section V discusses implications for deployment.

II. Related Work

This section reviews four research threads relevant to our contribution: foundation models for affective inference, cross-modal disagreement in multimodal sentiment analysis, uncertainty estimation and internal-state interpretation, and empa-

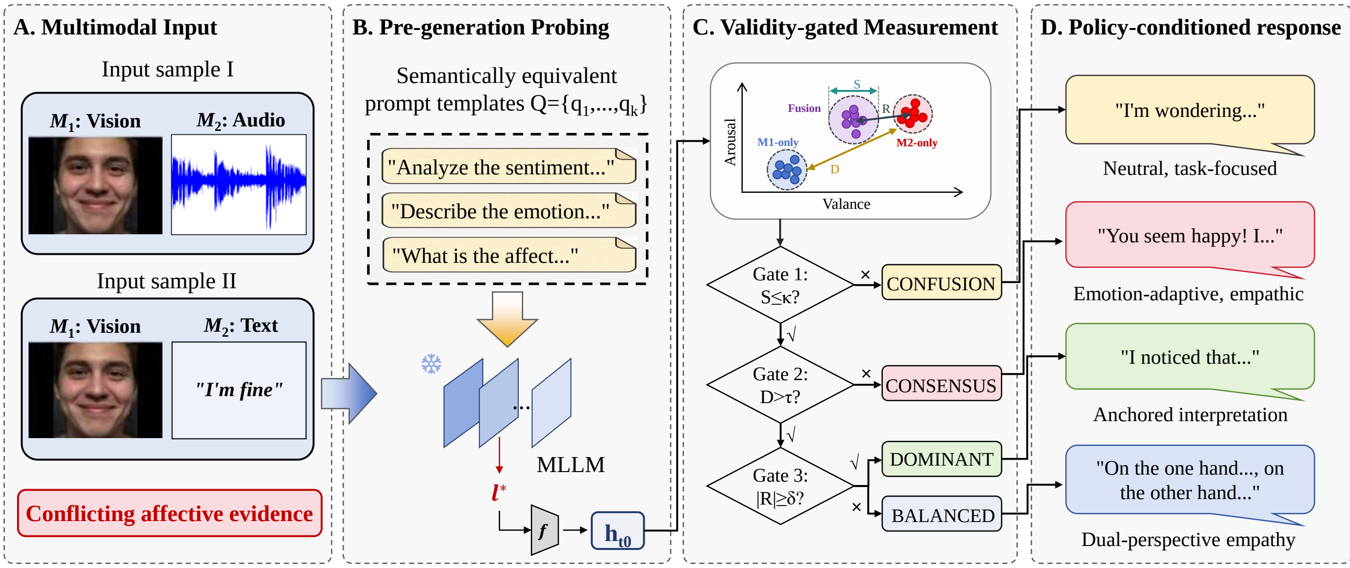

Fig. 1. Pre-generation auditing for affective modality conflict. (a) A single input can present conflicting affective evidence across modalities. (b) We probe a fixed pre-generation interface t_0 by applying semantically equivalent prompt templates $\mathcal{Q}$ and reading the hidden state at the last conditioning token of a frozen backbone. (c) Validity-gated measurement summarizes repeatability $\mathcal{S}$, split strength $\mathcal{D}$, and signed arbitration $\mathcal{R}$ (with $\mathcal{R}$ interpreted only on split-valid samples). (d) The resulting four-state taxonomy, comprising Consensus, Balanced, Dominant, and Confusion, provides an auditable signal for policy-conditioned responses.

thetic response generation. For each thread, we identify the gap that motivates our pre-generation measurement framework.

A. Foundation Models for Affective Inference

Recent multimodal foundation models extend large language models with visual and audio perception, enabling instruction-following interaction over heterogeneous inputs. Vision–language models such as LLaVA [7], BLIP-2 [8], and Qwen-VL [9] are trained through multimodal instruction tuning and evaluated on tasks including visual question answering and dialogue grounding. Audio–language and omni-modal variants further incorporate speech signals, supporting applications that require joint reasoning over facial expressions, vocal prosody, and verbal content [10], [11].

These models are increasingly applied to affective tasks, where the model is prompted to infer emotion categories, sentiment polarity, or clinically relevant signals from multimodal inputs. Emotion-LLaMA [12] and AffectGPT [13] demonstrate that instruction-tuned models can perform multimodal emotion recognition and generate affective explanations. However, evaluation in this line of work is predominantly focused on outputs: systems are judged by the accuracy of predicted labels, the coherence of generated explanations, or the appropriateness of supportive responses.

This evaluation paradigm does not directly expose how the model integrates potentially conflicting evidence before committing to a response. Once autoregressive decoding begins, the generated text reflects the chosen trajectory rather than the alternatives that were available at the moment of commitment. Research on rationale faithfulness has shown that post-hoc explanations produced by language models can be coherent yet structurally decoupled from the actual decision process [14], [15]. This limitation motivates measurement approaches anchored at the pre-generation interface, where modality-conditioned evidence has been encoded but no content token has yet been produced.

B. Cross-Modal Disagreement in Multimodal Sentiment Analysis

When confronted with modality conflicts, ambiguity, or insufficient information, large models exhibit a pronounced tendency toward overconfidence, generating coherent and assertive responses rather than explicitly acknowledging their uncertainty [16], [17]. In sentiment analysis contexts, this preference compels MLLMs to enforce logical closure upon contradictory affective signals (e.g., sarcasm, verbal-nonverbal inconsistency), yielding high-confidence yet logically erroneous sentiment judgments that amplify the risk of misleading decisions. Multimodal sentiment analysis benchmarks—including CMU-MOSI [18], CMU-MOSEI [19], and CH-SIMS [20]—contain numerous instances where verbal content and nonverbal expression are weakly aligned or mutually contradictory. Han et al. further established the specialized conflict benchmark dataset CA-MER [21], corroborating the ubiquity of affective modality conflicts in this domain. Wang et al. [24] demonstrated significant performance degradation when models encounter such conflicts, substantiating that affective modality conflicts increase model uncertainty and elevate decision error risks.

Current mainstream approaches predominantly treat affective modality conflicts as problems requiring resolution, employing cross-modal fusion or weight allocation mechanisms to suppress non-dominant modalities and obtain unified prediction labels [25], [26]. Recent research has begun to emphasize the significance of conflicting components: TriagedMSA [27] introduced a pre-screening network to detect sentiment polarity divergence, while Pang et al [28]. demonstrated the utility of leveraging conflict components in complex sentiment analysis through a multi-layer conflict-aware network. These methods confirm that disagreement constitutes not merely noise but a signal capable of enhancing downstream prediction performance; however, such studies nevertheless subject extracted conflict information to secondary fusion to construct improved predictors.

The present study adopts a distinct stance. Rather than exploiting conflict to refine fusion, we conceptualize conflict as a "structural event antecedent to generation," whose interpretation depends upon a well-defined domain of validity. This distinction differentiates our contribution from approaches that resolve disagreement through architectural innovation: we do not propose novel fusion mechanisms, but rather introduce a measurement protocol to characterize the integrated state of existing generative models at the precise moment of decision-making.

C. Uncertainty Estimation and Internal-State Interpretation

Uncertainty estimation for Large Language Models (LLMs) has advanced through the development of metrics based on output consistency and semantic stability. The semantic entropy approach [29] estimates epistemic uncertainty by aggregating variations across sampled outputs. Similarly, the self-consistency method [30] identifies unreliable predictions by examining the agreement among multiple decoding paths. Verbalized confidence elicitation [31], [32] prompts models to express uncertainty through natural language, although these expressions may not faithfully reflect the internal states of the model.

Research by Tomov et al. [34] indicates that traditional uncertainty estimation methods become ineffective in ambiguous scenarios, which include instances of emotional conflict. Furthermore, conventional detection methods struggle to effectively identify emotional modality conflicts. This limitation arises because traditional approaches generally assume that uncertainty stems from a lack of information or inherent vagueness, whereas emotional modality conflict manifests as dissonance resulting from the coexistence of multiple sharp and mutually exclusive posterior logics [33].

To uncover the internal mechanisms of models when they process conflicting information, researchers employ probing and readout techniques to analyze intermediate representations. Existing studies have demonstrated that the hidden layers of LLMs encode richer semantic and emotional information than the final outputs, forming a distinct internal emotional geometry [35], [36]. These findings provide more effective avenues for detecting emotional modality conflicts. Although mapping internal states to the vocabulary space, such as through the Logit Lens method, can reveal emotional tendencies during reasoning, this approach is susceptible to word frequency bias and alignment contamination, which results in readout deviations [37]. While frameworks such as Tuned Lens [37] and Patchscopes [38] have made breakthroughs in interpretability, ensuring the reliability and reproducibility of internal state readouts in complex environments characterized by emotional modality conflict remains a significant difficulty in current research. Consequently, the design of an explicit detection metric that is sensitive to emotional modality conflicts, robust against noise, and compliant with falsifiable standards has become a pivotal challenge for improving the robustness of multimodal sentiment analysis.

Output-level uncertainty and self-consistency methods operate downstream of generation, where decoding stochasticity is entangled with multimodal disagreement. Our readout targets the pre-generation interface t_0, where the model has not yet committed to a token trajectory. This makes template-sensitivity S and split/arbitration geometry $(\mathcal{D}, \mathcal{R})$ observable without conflating them with generation noise. The framework provides an upstream audit signal complementary to output-level metrics.

D. Empathetic Response Generation

A parallel research direction focuses on generating responses that are emotionally appropriate and supportive. Empathetic dialogue systems aim to recognize user emotional states and produce responses that acknowledge, validate, or address those states [39], [40]. Recent work integrates large language models into this paradigm, using instruction tuning or prompting strategies to elicit empathetic behavior [41], [42].

In mental-health-facing applications, the stakes of affective inference are elevated. Surveys on large language models in mental health emphasize both the opportunities and the risks, including the potential harm of overconfident or dismissive responses when users are in fragile states [43], [44]. Systems that commit to a single affective interpretation without detecting underlying conflict may inadvertently invalidate user experiences or miss signals of distress masked by social presentation.

Our measurement framework is complementary to response-generation strategies. By exposing the pre-generation decision state as an auditable interface, the framework supports downstream policies that can condition response behavior on whether the model's internal evidence is in consensus, in structural conflict, or in a non-repeatable regime. This enables interaction designs that prefer clarification or disclosure over blind commitment when the available evidence does not support a unitary affective claim.

III. Method

We formalize affective modality conflict as a structural event observable at a fixed pre-generation interface. The measurement framework consists of four coordinated components: (i) an observation interface at the last conditioning token t_0, causally upstream of any decoded content; (ii) an affective coordinate scale calibrated on a consistency domain and frozen thereafter; (iii) repeatability estimation via equivalent prompt templates; and (iv) validity gating that restricts structural interpretation to samples passing both repeatability and split-activation thresholds. The key design principle is that interpretation is validity-bounded: we explicitly delineate a domain within which structural claims are admissible, and report outcomes outside this domain as inadmissible for structural interpretation. Unless stated otherwise, all analyses involving the arbitration bias $\mathcal{R}$ are restricted to the split-valid domain $\mathcal{V}_{\mathrm{split}}$.

A. Problem Formulation and Observation Interface

We consider a bimodal evidence pair $X = (X_{M_1}, X_{M_2})$ where M_1 and M_2 denote two input modalities. The framework accommodates different modality combinations through two protocol instantiations: the *Omni protocol* with $(M_1, M_2) = (V, A)$, where V denotes visual input (video frames) and A denotes audio input encoding prosody and paralinguistic cues; and the *VL protocol* with $(M_1, M_2) = (V, T)$, where T denotes a text transcript obtained via automatic speech recognition.

For each sample, we evaluate three modality-availability conditions $m \in \{1, 2, 12\}$, where $m = 1$ exposes only M_1, $m = 2$ exposes only M_2, and $m = 12$ exposes both modalities jointly. This design enables comparison between unimodal and joint representations at the same observation interface.

The observation time t_0 is operationally defined as the position of the last token in the serialized conditioning sequence during the prefill step, before the first content token is generated. For a model with layers indexed by $\ell \in \{1, \ldots, L\}$, let $\mathbf{h}_{t_0}^{(\ell)}(m, q) \in \mathbb{R}^{d_h}$ denote the hidden state at layer ℓ and position t_0, extracted under modality condition m and prompt template q. A single layer $\ell^\star$ is selected on a calibration holdout by maximizing the coefficient of determination R^2 for affective coordinate prediction, and frozen for all evaluations. When the layer index is unambiguous, we write $\mathbf{h}_{t_0}(m, q) \equiv \mathbf{h}_{t_0}^{(\ell^\star)}(m, q)$.

Rationale for bimodal protocols. We do not evaluate a trimodal protocol (V, A, T) with all three modalities presented simultaneously. For Omni-modal models, the audio stream A already encodes substantial linguistic content through implicit speech recognition; presenting an explicit ASR transcript T alongside A introduces redundant and potentially conflicting textual signals, confounding the measurement of genuine cross-modal affective disagreement. The bimodal protocols isolate interpretable conflict axes: Omni (V, A) captures visual-prosodic disagreement, while VL (V, T) captures visual-linguistic disagreement. Extension to higher-order split geometry among three or more non-redundant modalities is deferred to future work.

B. Affective Coordinate Scale

We map the high-dimensional hidden state $\mathbf{h}_{t_0}$ into a d-dimensional affective coordinate system via a fixed readout function $f : \mathbb{R}^{d_h} \to \mathbb{R}^d$:

$$\hat{\mathbf{a}}_m(q) = f\big(\mathbf{h}_{t_0}(m, q)\big) \in \mathbb{R}^d. \tag{1}$$

The target dimension d depends on the available annotations: $d = 2$ for valence–arousal coordinates [22], and $d = 1$ for scalar sentiment intensity.

The readout function is calibrated on a designated *consistency domain* $\mathcal{B}$. This domain is constructed from samples at conflict level $L1$, where $L1$ is defined as having cross-modal inconsistency $c < 0.2$ on the normalized annotation scale (see Section IV-A for the full stratification procedure). The $L1$ criterion ensures that $\mathcal{B}$ contains samples with closely aligned modality annotations, providing a reliable scale-setting domain. We partition $\mathcal{B}$ into disjoint calibration and holdout subsets: $\mathcal{B} = \mathcal{B}_{\mathrm{cal}} \cup \mathcal{B}_{\mathrm{hold}}$. All components of the measurement protocol, including the readout function, layer selection, and validity thresholds, are determined on $\mathcal{B}$ and subsequently frozen for evaluation on conflict domains $L2$–$L4$. The consistency domain contains $|\mathcal{B}| = 1{,}800$ samples, with 80% in $\mathcal{B}_{\mathrm{cal}}$ and 20% in $\mathcal{B}_{\mathrm{hold}}$.

We instantiate f as a linear ridge regressor $f(\mathbf{h}) = \mathbf{W}\mathbf{h} + \mathbf{b}$, with parameters estimated by minimizing:

$$(\mathbf{W}^\star, \mathbf{b}^\star) = \arg\min_{\mathbf{W}, \mathbf{b}} \frac{1}{|\mathcal{B}_{\mathrm{cal}}|} \sum_{i \in \mathcal{B}_{\mathrm{cal}}} \big\|\mathbf{W}\mathbf{h}_{t_0, i} + \mathbf{b} - \mathbf{a}_i\big\|_2^2 + \lambda \|\mathbf{W}\|_F^2, \tag{2}$$

where $\mathbf{a}_i \in \mathbb{R}^d$ is the annotated affective coordinate and $\lambda > 0$ is selected on $\mathcal{B}_{\mathrm{hold}}$ to maximize R^2.

The readout f serves as a calibrated measurement scale rather than a predictive model. Its role is to provide unit and direction calibration within $\mathcal{B}$. Absolute coordinate values $\hat{\mathbf{a}}_m(q)$ are not assumed to have ground-truth semantics outside $\mathcal{B}$; however, relative geometric relationships, including the separation between unimodal summaries and the proximity ratio underlying $\mathcal{R}$, depend on f being a stable linear map in the calibrated neighborhood induced by $\mathcal{B}$ and template perturbations, with interpretation restricted by the gating protocol rather than assumed globally. This scope clarifies that the structural metrics $\mathcal{D}$ and $\mathcal{R}$ measure geometric relationships in the readout space, with interpretation validity enforced by the gating protocol rather than by global readout accuracy.

C. Repeatability Estimation

A central component of the framework is the estimation of measurement repeatability. We treat prompt variation not as a nuisance to be eliminated, but as a controlled perturbation that reveals the stability of the pre-generation state.

Semantically equivalent prompt templates are used to probe measurement repeatability. We fix the number of templates to $K = 5$ in all experiments. This value is selected by a preliminary template-count study that evaluates stability as K increases up to an upper bound of 16 candidate templates; after selection, the template set $\mathcal{Q} = \{q_1, \ldots, q_K\}$ is frozen and no template sampling is performed during evaluation. All K templates are semantically identical in intent and measurement target, differing only in surface form. Specifically, the templates satisfy four conditions: all templates convey identical evidence content and measurement intent; all templates specify identical modality-availability declarations for each condition m; all templates maintain identical serialization structure such that t_0 corresponds to the same interface position; and no template introduces auxiliary instructions that alter the measurement target. Templates are verified by human inspection prior to any evaluation.

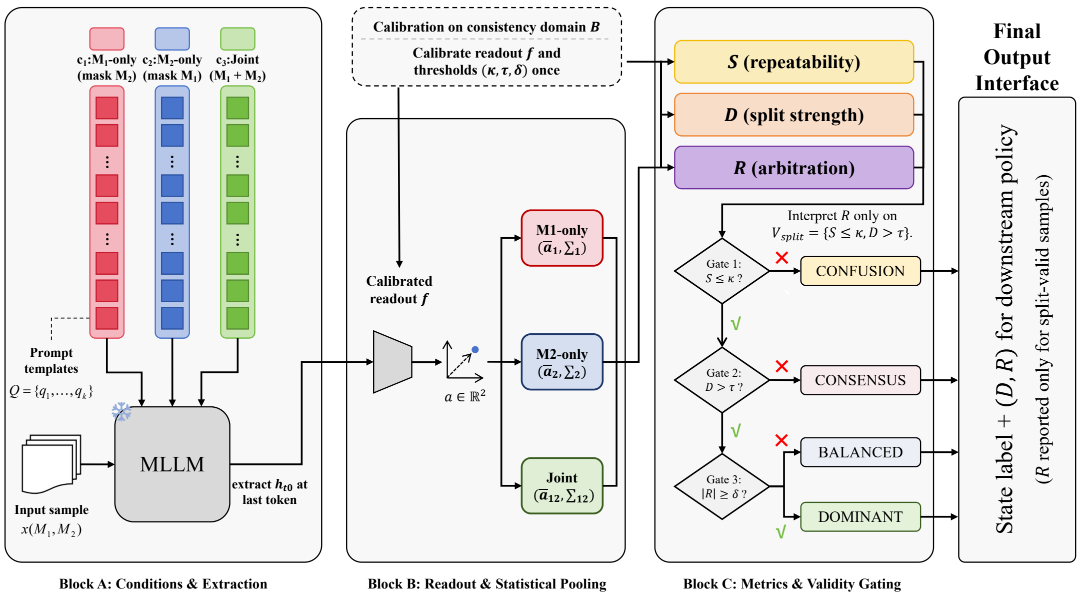

Fig. 2. Architecture of the validity-gated readout protocol in the Audit Module. Block A extracts K pre-generation states $\{\mathbf{h}_{t_0}(q_k, m)\}$ under three conditions (M_1-only, M_2-only, and Joint) using a frozen backbone and a prompt template set $\mathcal{Q}$. Block B applies a calibrated readout f and performs statistical pooling to estimate $(\bar{\mathbf{a}}_1, \mathbf{\Sigma}_1)$, $(\bar{\mathbf{a}}_2, \mathbf{\Sigma}_2)$, and $(\bar{\mathbf{a}}_{12}, \mathbf{\Sigma}_{12})$. Block C computes S, $\mathcal{D}$, and $\mathcal{R}$ (Eqs. 5–8) and applies a three-step gating logic with thresholds (κ, τ, δ) calibrated once on a consistency domain $\mathcal{B}$ and frozen for evaluation, yielding a state label and auditable outputs for downstream policies.

For each modality condition $m \in \{1, 2, 12\}$, repeated measurements over $\mathcal{Q}$ yield an ensemble $\{\hat{\mathbf{a}}_m(q)\}_{q\in\mathcal{Q}}$. We compute the template-averaged mean and sample covariance:

$$\bar{\mathbf{a}}_m = \frac{1}{K}\sum_{q\in\mathcal{Q}} \hat{\mathbf{a}}_m(q), \tag{3}$$

$$\mathbf{\Sigma}_m = \frac{1}{K-1}\sum_{q\in\mathcal{Q}} \big(\hat{\mathbf{a}}_m(q) - \bar{\mathbf{a}}_m\big)\big(\hat{\mathbf{a}}_m(q) - \bar{\mathbf{a}}_m\big)^\top. \tag{4}$$

The covariance $\mathbf{\Sigma}_m$ quantifies the repeatability bandwidth: it measures how much the readout varies when only the prompt phrasing changes while evidence content remains fixed. The dispersion estimates assume template exchangeability and local linearity of f over the range of hidden-state variation induced by template perturbation. We add a regularization term $\epsilon\mathbf{I}$ with $\epsilon = 10^{-6}$ to ensure numerical stability.

D. Validity Gating and State Taxonomy

We define explicit validity gates that determine which samples admit structural interpretation, based on two criteria: measurement repeatability and split activation.

Stability gate. We aggregate unimodal dispersions into a scalar stability statistic:

$$S = \mathrm{tr}(\mathbf{\Sigma}_1) + \mathrm{tr}(\mathbf{\Sigma}_2). \tag{5}$$

The stability gate is defined by a threshold κ: $\mathcal{V}_{\text{stab}} \triangleq \{S \le \kappa\}$. Samples with $S > \kappa$ exhibit high template-induced dispersion, indicating that the modality-conditional summaries are not repeatably localized; structural claims about their separation are inadmissible.

Split strength. For samples passing the stability gate, we quantify the separation between unimodal summaries using a dispersion-normalized distance:

$$\mathcal{D} = \sqrt{(\bar{\mathbf{a}}_1 - \bar{\mathbf{a}}_2)^\top(\mathbf{\Sigma}_1 + \mathbf{\Sigma}_2 + \epsilon\mathbf{I})^{-1}(\bar{\mathbf{a}}_1 - \bar{\mathbf{a}}_2)}. \tag{6}$$

This metric has the form of a Mahalanobis-like distance: when unimodal means are well-separated relative to their combined dispersion, $\mathcal{D}$ is large; when the separation is comparable to or smaller than the dispersion bandwidth, $\mathcal{D}$ is small. The split-valid domain is defined by a threshold τ:

$$\mathcal{V}_{\text{split}} \triangleq \mathcal{V}_{\text{stab}} \cap \{\mathcal{D} > \tau\} = \{S \le \kappa\} \cap \{\mathcal{D} > \tau\}. \tag{7}$$

Threshold calibration. Both thresholds are calibrated on $\mathcal{B}$ using fixed quantile rules: $\kappa = \hat{Q}_S(q_\kappa; \mathcal{B})$ and $\tau = \hat{Q}_{\mathcal{D}}(q_\tau; \mathcal{B})$, where $\hat{Q}_Z(q; \mathcal{B})$ is the empirical q-quantile. We use $q_\kappa = q_\tau = 0.95$ throughout. Once calibrated, thresholds are frozen for evaluation. Cross-model comparisons use the normalized statistic $\tilde{\mathcal{D}} = \mathcal{D}/\tau$.

Arbitration bias. For samples within $\mathcal{V}_{\text{split}}$, we quantify arbitration bias by measuring the geometric proximity of the joint summary $\bar{\mathbf{a}}_{12}$ to the two unimodal endpoints:

$$\mathcal{R} = \frac{\|\bar{\mathbf{a}}_{12} - \bar{\mathbf{a}}_2\|_2 - \|\bar{\mathbf{a}}_{12} - \bar{\mathbf{a}}_1\|_2}{\|\bar{\mathbf{a}}_1 - \bar{\mathbf{a}}_2\|_2 + \epsilon} \in [-1, 1]. \tag{8}$$

When $\mathcal{R} > 0$, the joint summary is closer to $\bar{\mathbf{a}}_1$ (modality M_1); when $\mathcal{R} < 0$, it is closer to $\bar{\mathbf{a}}_2$. The arbitration bias $\mathcal{R}$ is assigned semantic interpretation only within $\mathcal{V}_{\text{split}}$; outside this domain, $\mathcal{R}$ is reported but not assigned directional semantics.

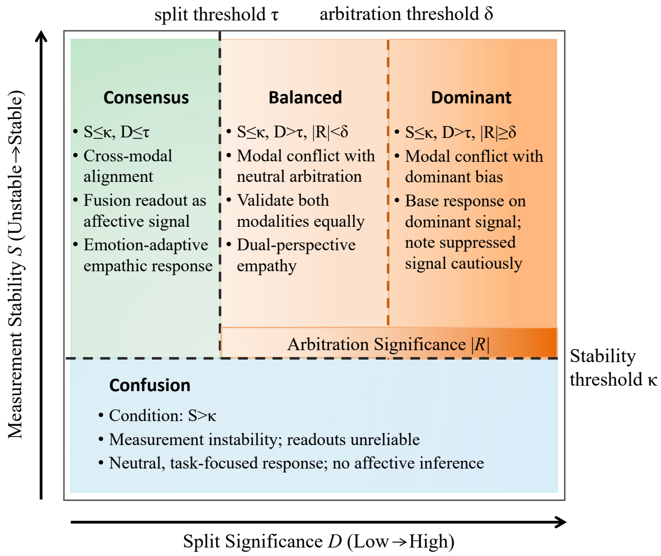

Fig. 3. Architecture of the Response Module. The module receives the taxonomic state label and geometric coordinates from the Audit Module. Based on the state label, it selects the corresponding policy (A–D) and constructs a policy-conditioned prompt that enforces compliance constraints during response generation.

Four-state taxonomy. We compose the validity gates into a mutually exclusive and exhaustive taxonomy. For a sample with statistics $(S, \mathcal{D}, \mathcal{R})$:

$$\textsc{Confusion}: \quad S > \kappa, \tag{9}$$

$$\textsc{Consensus}: \quad S \leq \kappa \text{ and } \mathcal{D} \leq \tau, \tag{10}$$

$$\textsc{Balanced}: \quad S \leq \kappa,\ \mathcal{D} > \tau,\ \text{and } |\mathcal{R}| < \delta, \tag{11}$$

$$\textsc{Dominant}: \quad S \leq \kappa,\ \mathcal{D} > \tau,\ \text{and } |\mathcal{R}| \geq \delta, \tag{12}$$

where δ is a significance threshold for arbitration bias. Confusion indicates non-repeatable measurement where no structural claim is admissible. Consensus indicates repeatable readouts without significant separation. Balanced indicates a significant split without systematic proximity to either endpoint. Dominant indicates a significant split with directional bias toward one modality.

To distinguish Balanced from Dominant, we test whether $|\mathcal{R}|$ exceeds a template-induced uncertainty band. We approximate the standard error of $\mathcal{R}$ via first-order error propagation [23]. Let $g : \mathbb{R}^{3d} \to \mathbb{R}$ denote Eq. (8) as a function of the concatenated mean vector $\mathbf{x} = [\bar{\mathbf{a}}_1^\top, \bar{\mathbf{a}}_2^\top, \bar{\mathbf{a}}_{12}^\top]^\top$. The standard error is

$$\mathrm{SE}(\mathcal{R}) = \sqrt{\nabla g(\mathbf{x})^\top \mathrm{Cov}(\mathbf{x}) \nabla g(\mathbf{x})}, \tag{13}$$

with the covariance matrix defined as

$$\mathrm{Cov}(\mathbf{x}) = \frac{1}{|\mathcal{Q}|} \begin{bmatrix} \mathbf{\Sigma}_1 & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{\Sigma}_2 & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{\Sigma}_{12} \end{bmatrix}. \tag{14}$$

The significance threshold is $\delta = z_{1-\gamma/2} \cdot \mathrm{SE}(\mathcal{R})$, where $z_{1-\gamma/2}$ is the standard normal quantile. We use $\gamma = 0.05$, corresponding to $z_{0.975} = 1.96$.

E. Scope and Interpretation Constraints

The framework measures geometric relationships in the pre-generation representation space; it does not adjudicate which modality carries the "correct" affective signal. In cases of genuine conflict, there may be no single correct interpretation. The four-state taxonomy describes the model's internal state, not a ground-truth affective label.

The affective coordinate scale is a calibrated local chart, not a universal semantic guarantee. On evaluation domains with high conflict, we do not assume the same linear chart remains globally valid. Degradation of linear readout fit at high conflict levels is treated as boundary evidence rather than framework failure (see Section IV-B). The taxonomy labels describe geometric configurations of model internal states and must not be interpreted as user diagnoses or clinical determinations.

Interpretation domain switching. The joint summary $\bar{\mathbf{a}}_{12}$ is interpreted as an affective coordinate only when the linear readout remains adequate (empirically assessed on $\mathcal{B}$ and low-conflict levels). Under high conflict, joint R^2 degrades (Section IV-B); we therefore stop using $\bar{\mathbf{a}}_{12}$ for coordinate claims and switch to *relational auditing*: $\mathcal{D}$ measures unimodal separation ($\bar{\mathbf{a}}_1$ vs. $\bar{\mathbf{a}}_2$), and $\mathcal{R}$ is treated as a proximity-based arbitration index of the joint summary relative to the two endpoints. In this regime, the protocol outputs a routed state label rather than a regressed affective coordinate.

Fig. 2 summarizes the architecture of the Audit Module described above.

F. Transition to Action: Measurement-to-Policy

The four-state taxonomy provides a geometric characterization of the pre-generation integration state; however, geometric configurations alone do not dictate interaction protocols. To bridge this gap, we introduce the Response Module, where we establish a state-conditioned policy for each identified state. Specifically,each state $\in$ $\{\text{Confusion}, \text{Consensus}, \text{Balanced}, \text{Dominant}\}$ activates a distinct response policy, designed to align the system's communicative stance with the epistemic validity of its internal measurements.Operationally, these policies are realized by prepending strategy-specific system instructions to the generation prompt.Fig. 3 illustrates the architecture of the Response Module.

Policy A (Dominant split-valid). On $\mathcal{V}_{\text{split}}$ with $c_{\mathcal{R}}$ =Dominant, the system generates a response that (i) discloses that modalities encode incompatible affective evidence, (ii) anchors its tentative affective interpretation to the closer endpoint (sign of $\mathcal{R}$), and (iii) explicitly solicits disambiguating information about the suppressed channel. The policy forbids single-valued affect assertions that ignore the farther endpoint.

Policy B (Balanced split-valid). On $\mathcal{V}_{\text{split}}$ with $c_{\mathcal{R}}$ =Balanced, the system produces a two-branch acknowledgment that preserves both endpoints $\bar{\mathbf{a}}_1$ and $\bar{\mathbf{a}}_2$ as concurrent hypotheses, followed by a clarification question designed to separate which channel reflects the user's intended message.

Policy C (Consensus). On $S \le \kappa$ and $\mathcal{D} \le \tau$, the system uses the joint readout $\bar{\mathbf{a}}_{12}$ as a stable affective coordinate for tone calibration (for example, choosing between supportive and exploratory phrasing), without introducing conflict disclosure. **Policy D (Confusion).** On $S > \kappa$, the system refrains from affective inference and instead asks for additional information (or rephrasing) while maintaining a neutral tone. Only non-affective task content is addressed.

IV. Experiments

We evaluate two predictions: (i) split strength $\mathcal{D}$ increases with cross-modal inconsistency, and (ii) within the split-valid domain, arbitration bias $\mathcal{R}$ drifts systematically as $\mathcal{D}$ increases, with direction varying by model.

A. Experimental Setup

Datasets and conflict stratification. We use CMU-MOSI [18], CMU-MOSEI [19], and CH-SIMS v2 [20]. Raw annotations are normalized to $[-1, 1]$. For CH-SIMS v2, the dataset provides independent human annotations (y_V, y_A, y_T) per modality [20]. For CMU-MOSI and CMU-MOSEI, we use the dataset-provided unimodal sentiment annotations for each modality [18], [19]. Cross-modal inconsistency is defined as $c = |y_V - y_A|$ (Omni) or $c = |y_V - y_T|$ (VL). Samples are stratified into four levels: $L1$ ($c < 0.2$), $L2$ ($0.2 \le c < 0.6$), $L3$ ($0.6 \le c < 1.0$), and $L4$ ($c \ge 1.0$). The consistency domain $\mathcal{B}$ comprises $L1$ samples; $L2$–$L4$ form the evaluation domain.

Models. For the Omni protocol (V, A): Qwen2.5-Omni-7B [11]. For the VL protocol (V, T): Qwen3-VL-8B [45] and InternVL3_5-8B [46].

Protocol parameters. Template count $K = 5$, threshold quantiles $q_\kappa = q_\tau = 0.95$, significance level $\gamma = 0.05$, regularization $\epsilon = 10^{-6}$. Cross-model comparisons use normalized split strength $\tilde{\mathcal{D}} = \mathcal{D}/\tau$.

Computational overhead. The protocol requires $3|\mathcal{Q}| = 15$ forward passes per sample. However, actual overhead is substantially lower than 15 complete inference cycles: the protocol requires only prefill to extract $\mathbf{h}_{t_0}$, without autoregressive generation, and modern engines (e.g., vLLM) support parallel prefill. On a single A100-80GB with Qwen2.5-Omni-7B, 15-pass prefill completes in 4233ms—37.9% faster than full response generation (5836ms). The prefill KV-cache can be reused by the Response Module, reducing Stage II latency by 44.9%.

B. Calibration and Protocol Validation

Template count selection. Fig. 4 reports normalized stability gain $(1 - \sigma_{\text{norm}})$ as K increases. We apply the Kneedle algorithm [49], which identifies the point maximally distant from the line connecting curve endpoints after normalization. The knee occurs at $K = 5$, where template perturbation has sufficiently sampled the hidden state's sensitivity to paraphrase variants. The rapid convergence suggests that pragmatic factors influencing $\mathbf{h}_{t_0}$ lie in a low-rank subspace rather than a high-dimensional random distribution.

Layer selection. Fig. 5 shows layer-wise R^2 for Qwen2.5-Omni-7B. Readout accuracy peaks at $\ell^\star = 21$ with $R^2 =$ 0.793, rather than the final layer, indicating that affective summarization completes before generation commitment and is subsequently diluted by task-irrelevant information. The V-only, A-only, and VA-joint curves rise in near-lockstep through layer 20, suggesting a shared affective abstraction pathway. The joint condition exhibits the steepest post-peak decline, consistent with fusion representations encoding generation-oriented formatting beyond pure affective semantics.

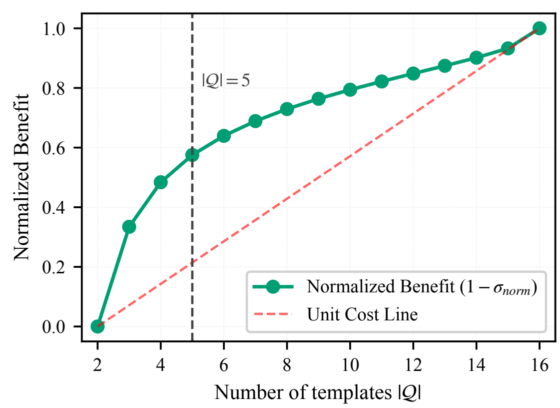

Fig. 4. Template count selection via Kneedle algorithm. The knee at $K = 5$ marks optimal cost-benefit balance.

Calibrated thresholds. On $\mathcal{B}$, we obtain $\kappa = 0.22$, $\tau = 0.311$ for Qwen2.5-Omni-7B; $\kappa = 0.22$, $\tau = 0.570$ for Qwen3-VL-8B; $\kappa = 0.29$, $\tau = 0.932$ for InternVL3_5-8B.

Readout validity and interpretation domain. Fig. 6 shows that unimodal readouts degrade with conflict (V: $0.78 \to 0.34$; A: $0.81 \to 0.32$ from $L1$ to $L4$), while the joint readout degrades more sharply (VA: $0.84 \to 0.22$). We therefore do not treat $\bar{\mathbf{a}}_{12}$ as an affective coordinate at high conflict. Instead, we use $\mathcal{D}$ and $\mathcal{R}$ as relational audit signals: $\mathcal{D}$ captures unimodal separation, and $\mathcal{R}$ indicates the joint summary's proximity to either endpoint. Fig. 7 shows monotonic increase of $\tilde{\mathcal{D}}$ with conflict dosage, supporting ordered split detection even when joint regression is unreliable. The output at high conflict is a routed state label, not a regressed coordinate.

Non-linear readout ablation. A two-layer MLP readout achieves $R^2 = 0.90$ on $\mathcal{B}$, a modest improvement over the linear baseline (0.80). However, layer selection becomes unstable: R^2 increases monotonically with depth and peaks at the final layer, suggesting the MLP exploits high-dimensional cues beyond affective semantics. We retain linear readout as a principled trade-off between interpretability and fit.

C. Split Response to Conflict Dosage

Fig. 7 shows a clear dose-ordered increase in normalized split strength $\tilde{\mathcal{D}}$ across both models, with the high-conflict band crossing the operational split threshold ($\tilde{\mathcal{D}} > 1$). This indicates that a readout calibrated on the low-conflict domain preserves ordered unimodal separation under distributional shift, enabling conflict detection even when joint coordinate regression is unreliable. Qwen2.5-Omni-7B exhibits greater dispersion at high conflict, implying more heterogeneous split responses under severe inconsistency.

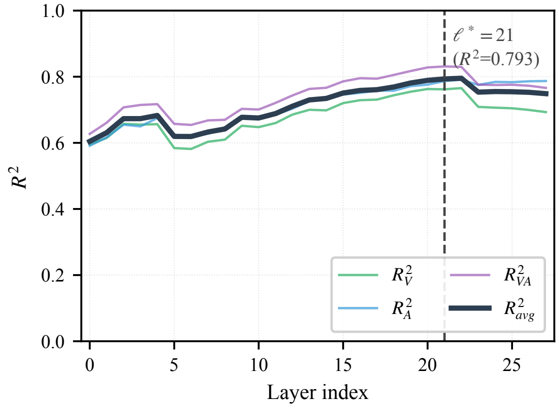

Fig. 5. Layer-wise R^2 for Qwen2.5-Omni-7B. Peak at $\ell^\star = 21$ indicates affective encoding completes mid-network.

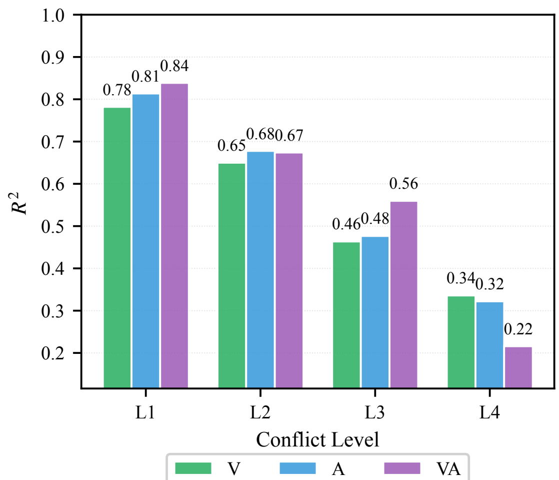

Fig. 6. R^2 by conflict level (Qwen2.5-Omni-7B).

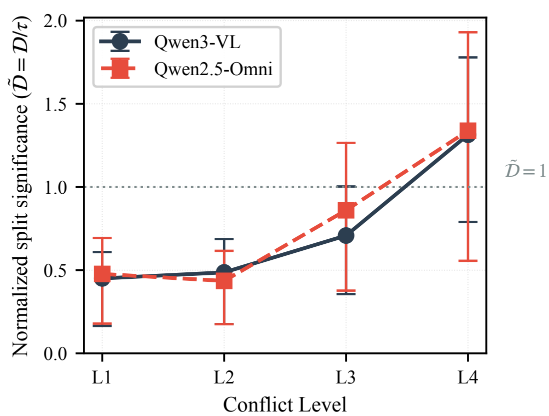

Fig. 7. Normalized split strength by conflict level.

Fig. 8 visualizes the pre-generation geometry in the $\mathcal{D}$–$\mathcal{R}$ plane for Qwen2.5-Omni-7B on stability-passing samples. $\mathcal{D}$ indexes the separation between the modality-conditional summaries, and $\mathcal{R}$ indexes the relative proximity of the joint summary to Visual ($\mathcal{R} > 0$) versus Audio ($\mathcal{R} < 0$). Each point is color-coded by arbitration bias, with red indicating Visual-leaning fusion and blue indicating Audio-leaning fusion. The two insets provide an explicit geometric interpretation in the linear affective coordinate space: in the low-$\mathcal{D}$ inset, the unimodal readouts lie close to each other and the joint readout (VA in orange) sits between them, consistent with agreement-like fusion; in the high-$\mathcal{D}$ inset, the unimodal readouts are well separated and the joint readout collapses toward one endpoint, reflecting directional arbitration rather than midpoint interpolation. The figure annotates three interpretable regimes induced by the thresholds: Consensus ($\mathcal{D} \le \tau$), Balanced ($\mathcal{D} > \tau$, $|\mathcal{R}| < \delta$), and Dominant ($\mathcal{D} > \tau$, $|\mathcal{R}| \ge \delta$).

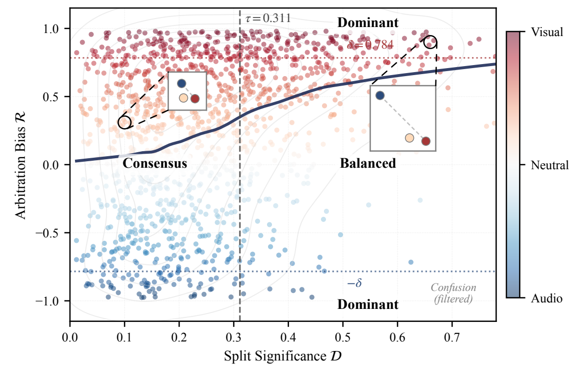

Fig. 8. $\mathcal{D}$–$\mathcal{R}$ geometry for Qwen2.5-Omni-7B.

Compared to the Qwen2.5-Omni-7B in Fig. 8, the VL models exhibit different arbitration signatures in Fig. 9. Qwen3-VL-8B exhibits a near-neutral arbitration profile: the conditional trend stays close to $\mathcal{R} = 0$ and split-valid points do not concentrate in a single dominant direction. In contrast, InternVL3_5-8B displays a strong Text-leaning profile, with split-valid samples predominantly falling at $\mathcal{R} < 0$ and the conditional trend decreasing with $\mathcal{D}$. The thresholds differ materially across models: InternVL3_5 uses a higher split threshold τ, implying a stricter criterion for declaring separation, while its smaller δ makes dominance assignment more sensitive once split is declared. Together, these plots establish model-specific arbitration signatures under an identical measurement protocol.

D. State Distribution Across Conflict Levels

Fig. 10 reports the four-state composition across conflict levels. Across both backbones, increasing inconsistency primarily manifests as a redistribution out of Consensus into split-valid states (Balanced + Dominant), indicating that higher conflict more frequently produces a resolvable pre-generation split. Confusion does not track conflict monotonically: it stays low and relatively stable for Qwen2.5-Omni but rises noticeably at $L4$ for Qwen3-VL, suggesting that high-conflict VL inputs more often trigger stability failure under template perturbations.

The overlaid $p_{\text{dom}|\text{split}}$ highlights distinct arbitration profiles. Qwen3-VL shows an upward drift in dominance among

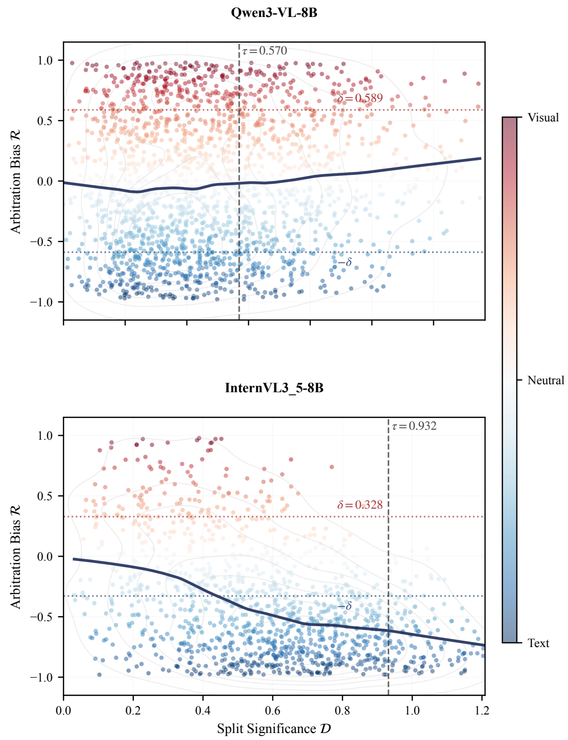

Fig. 9. $\mathcal{D}$–$\mathcal{R}$ geometry for VL models.

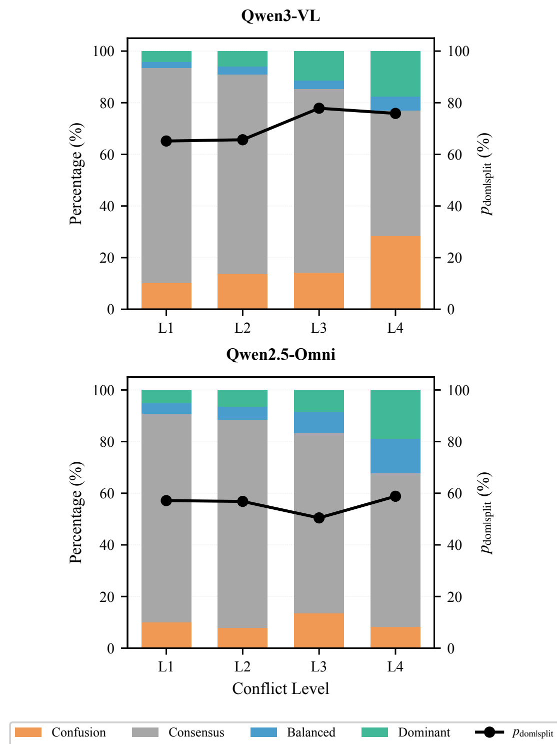

Fig. 10. Four-state distribution by conflict level.

split-valid samples as conflict increases, whereas Qwen2.5-Omni remains comparatively steady, indicating that dominance is not a deterministic function of conflict dosage. Notably, Balanced does not correspond to "moderate conflict": its prevalence is non-monotonic, consistent with an interpretation of "split established but arbitration near-neutral" rather than an intermediate severity band. In other words, conflict increases the frequency of split, while the direction of arbitration is a backbone signature.

E. Downstream Interaction Metrics and Setup

The Response Module evaluates empathetic response generation conditioned on pre-generation decision states. The evaluation employs the samples from the Audit Module evaluation domain $\mathcal{E}$ (Section IV-B), retaining their pre-computed taxonomic states (Confusion, Consensus, Balanced, Dominant) and geometric coordinates $(S, \mathcal{D}, \mathcal{R})$ derived from Section IV-D.

We employ a hybrid evaluation framework to assess both the system's adherence to state-conditioned strategies and the perceived quality of the interaction.

a) Adherence to State-Conditioned Policy: To quantify the system's fidelity to the prescribed response strategies, we employ an LLM-based automated evaluation mechanism (using Gemini-3-Flash). Instead of relying on generic quality scores, this metric specifically assesses whether the generated response structurally aligns with the logic constraints imposed by the detected decision state. The evaluator verifies distinct compliance criteria across different regimes: for samples marked as *Confusion*, it checks if the model successfully suspends affective assertions to prevent hallucination; for *Consensus* states, it assesses consistency with the detected polarity; for *Dominant* samples, it verifies if the response explicitly acknowledges the suppressed affective cues; and for *Balanced* cases, it measures the maintenance of an equitable dual-perspective discourse.

b) Human-Perceived Interaction Quality: Human evaluators assess the generated content across four dimensions: *Engagement*, *Understanding*, *Sympathy* and *Helpfulness*. Scoring employs a 5-point Likert scale, focusing on whether the pre-generation geometry and the enforced policy constraints (Section III-F) actually produce superior downstream interaction compared to blind generation.

F. Policy-Conditioned Generation Configurations

To isolate component contributions, we implement four generation configurations:

Base: Raw multimodal inputs with generic instruction, without access to Audit Module outputs.

Policy-Only: Inputs prompted with state-aware instructions corresponding to Policies A–D (Section III-F), but without provision of Audit Module decision metrics or the derived taxonomic states.

Signal-Guided: Inputs augmented with explicit Audit Module decision signals including taxonomic state labels, operating under generic instructions without policy-specific constraint enforcement.

Ours: Complete pipeline where $(S, \mathcal{D}, \mathcal{R})$ determines taxonomic state and enforces the corresponding policy (A, B, C, or D) with full compliance requirements.

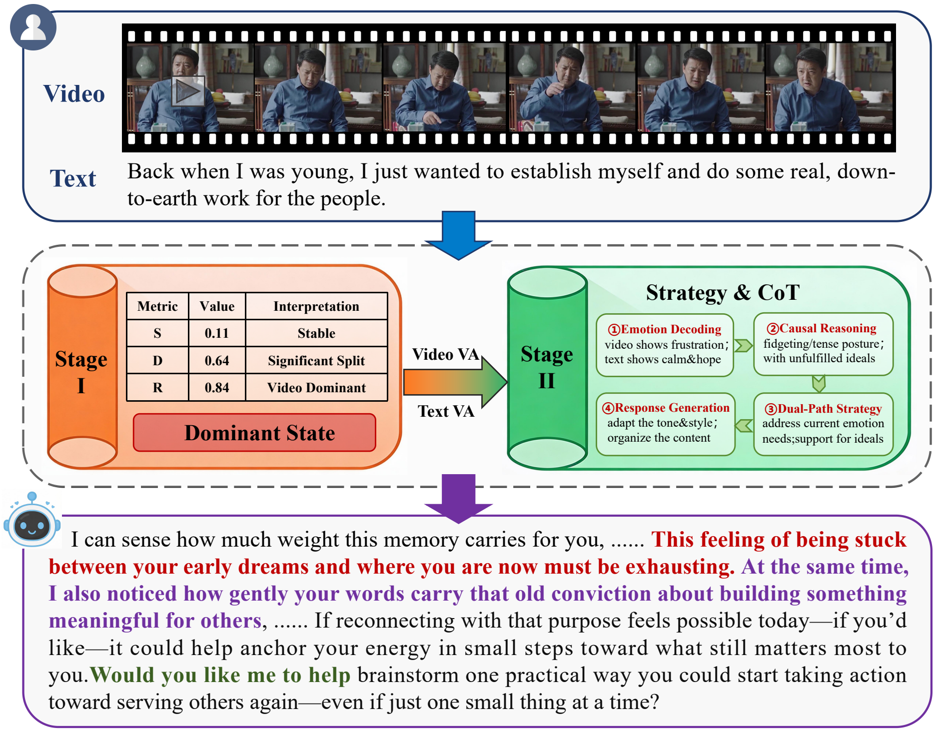

Fig. 11. End-to-end case study. Stage I detects Dominant state with visual bias; Stage II generates a policy-compliant response anchored to the dominant modality.

G. Behavioral Effects of Decision States

a) Instruction Adherence.: We first verify that models equipped with the proposed response policy can reliably follow the required instructions. As shown in Table I, both Qwen2.5-Omni and Qwen3-VL achieve high adherence scores, exceeding 0.90 on average. This result indicates that the strategy can be stably executed and does not introduce instruction-following failures.

b) Response-Level Effects.: Table II reports response-level evaluation results under different policy configurations. Introducing the empathy-oriented response strategy alone yields consistent but limited improvements across all models, suggesting that prompt-level conditioning primarily affects surface expression patterns.

In contrast, when Audit Module decision signals are available, conditioning responses on these signals leads to substantially different response behaviors. The magnitude of this change exceeds that obtained from response-level prompting alone, indicating that pre-generation decision information plays a central role in shaping downstream interaction behavior.

Combining Audit Module signals with the empathy-oriented strategy produces the highest overall scores, although the incremental gain over using Audit Module signals alone is comparatively small. This pattern suggests that the response strategy mainly serves to calibrate expression once the underlying decision state has been structurally constrained.

We emphasize that this stage evaluates observable response behavior under controlled policy conditioning. The reported metrics do not assess psychological impact or user outcomes, and are intended solely to demonstrate how internal decision-state signals translate into externally observable interaction patterns.

TABLE I
INSTRUCTION ADHERENCE OF MODELS THAT REQUIRE EXPLICIT STRATEGY COMPLIANCE. SCORES ARE NORMALIZED TO THE RANGE $[0, 1]$, WHERE HIGHER VALUES INDICATE BETTER ADHERENCE.

Model	Setting	Instruction Adherence ↑
Qwen2.5-Omni	Our Policy	0.9091
Qwen3-VL	Our Policy	0.9583

TABLE II
EMPATHETIC RESPONSE QUALITY UNDER DIFFERENT STRATEGY CONFIGURATIONS. SCORES ARE REPORTED ON A $[0, 20]$ SCALE USED CONSISTENTLY ACROSS ALL MODELS AND SETTINGS.

Model	Base	Policy-Only	Signal-Guided	Ours
Qwen2.5-Omni-7B	9.48	11.31	18.67	**18.72**
Qwen3-VL-8B	14.39	17.40	16.78	**18.08**
GPT-4o	11.98	12.94	–	–
Gemini-3-Pro	19.15	19.41	–	–

H. Case Analysis

Fig. 11 illustrates the end-to-end pipeline on a representative sample exhibiting cross-modal affective conflict. The multi-modal input presents tension between visual and linguistic evidence: video frames reveal somatic tension and fidgeting behaviors suggestive of frustration, while the verbal content

("Back when I was young, I just wanted to establish myself...") conveys calm idealism and composed self-reflection.

The Audit Module detects a DOMINANT state with the following validity-gated metrics: stability $S = 0.11$ (passing the repeatability gate $S \leq \kappa$), split strength $\mathcal{D} = 0.64$ (exceeding threshold τ), and arbitration bias $\mathcal{R} = +0.84$ (indicating $|\mathcal{R}| \geq \delta$). Geometrically, the joint representation $\bar{\mathbf{a}}_{12}$ lies significantly closer to the Visual endpoint than to the Text endpoint in the affective coordinate space, confirming systematic bias toward non-verbal cues.

Conditioned on this audit signal, the Response Module activates Policy A, generating a response that anchors affective interpretation to the visually detected frustration ($\mathcal{R} > 0$) while acknowledging the suppressed textual signal and soliciting clarification.

This case demonstrates complete measurement-to-policy traceability: the pre-generation configuration $(S, \mathcal{D}, \mathcal{R})$ maps to the Dominant state, which triggers a response grounded in the detected visual tension while preserving epistemic humility toward the contradictory verbal content.

I. Limitations

Scope of phenomenon captured. The current formulation addresses bimodal pairs at a single observation time t_0; it does not address intra-modal temporal inconsistency within a single modality. Extensions to three or more modalities require higher-order split geometry not covered here. Conflict levels $L1$–$L4$ are derived from annotation discrepancy and serve as a controlled evaluation axis, not a causal account of how conflict arises. The four-state taxonomy describes geometric configurations of model internal states and should not be interpreted as diagnostic labels.

High-conflict regime behavior. Under strong cross-modal inconsistency, the joint readout $\bar{\mathbf{a}}_{12}$ becomes unreliable for coordinate regression; the protocol therefore switches to state routing rather than coordinate prediction. A non-negligible fraction of samples falls into the Confusion state, where the framework abstains from split/arbitration interpretation.

V. CONCLUSION

We proposed a pre-generation measurement framework that treats multimodal affective conflict as a structural event observable at a fixed interface t_0—the last conditioning token before content generation begins. The protocol yields an auditable triplet $(S, \mathcal{D}, \mathcal{R})$ at t_0, enabling state routing at the pre-generation interface.

Across graded inconsistency, $\tilde{\mathcal{D}}$ exhibits a monotonic dose–response, indicating increasingly separable modality-conditional summaries as conflict intensifies. Within the split-valid regime, $\mathcal{R}$ exposes backbone-specific arbitration signatures (e.g., visual-leaning for Qwen2.5-Omni, text-leaning for InternVL3_5, near-neutral for Qwen3-VL), which are not predictable from conflict dosage alone.

At high conflict, we use the joint readout for routing rather than affective coordinate claims: the output is a four-state label (Confusion, Consensus, Balanced, Dominant) that controls downstream response policies. Rather than committing blindly to a single affective interpretation, systems can condition response behavior on what is structurally supported by the pre-generation state.

Future work includes extending the geometry beyond bimodal pairs and incorporating intra-modal temporal inconsistency. A second direction is to explore richer affective coordinate systems while preserving auditability at the pre-generation interface.

REFERENCES

[1] H. R. Lawrence *et al.*, "The opportunities and risks of large language models in mental health," *JMIR Mental Health*, vol. 11, no. 1, p. e59479, Jul. 2024, doi: 10.2196/59479.

[2] L. Shen *et al.*, "Enhancing emotion regulation in mental disorder treatment: An AIGC-based closed-loop music intervention system," *IEEE Trans. Affect. Comput.*, vol. 16, no. 3, pp. 2245–2260, 2025, doi: 10.1109/TAFFC.2025.3557873.

[3] R. D. Gupta, M. T. Hosain, M. F. Mridha, and S. U. Ahmed, "Multimodal programming in computer science with interactive assistance powered by large language model," in *Proc. Int. Conf. Hum.-Comput. Interact.*, Cham, Switzerland, 2025, pp. 59–69.

[4] A. Jacovi and Y. Goldberg, "Towards faithfully interpretable NLP systems: How should we define and evaluate faithfulness?" in *Proc. 58th Annu. Meeting Assoc. Comput. Linguistics*, 2020, pp. 4198–4205.

[5] D. Hazarika, R. Zimmermann, and S. Poria, "MISA: Modality-invariant and -specific representations for multimodal sentiment analysis," in *Proc. 28th ACM Int. Conf. Multimedia (MM)*, Seattle, WA, USA, 2020, pp. 1122–1131.

[6] Y.-H. H. Tsai *et al.*, "Multimodal transformer for unaligned multimodal language sequences," in *Proc. 57th Annu. Meeting Assoc. Comput. Linguistics (ACL)*, Florence, Italy, 2019, pp. 6558–6569.

[7] H. Liu *et al.*, "Visual instruction tuning," in *Adv. Neural Inf. Process. Syst.*, vol. 36, 2023, pp. 34892–34920.

[8] J. Li *et al.*, "BLIP-2: Bootstrapping language-image pre-training with frozen image encoders and large language models," in *Proc. 40th Int. Conf. Mach. Learn. (ICML)*, Honolulu, HI, USA, 2023, pp. 19730–19742.

[9] J. Bai *et al.*, "Qwen-VL: A versatile vision-language model for understanding, localization, text reading, and beyond," arXiv preprint arXiv:2308.12966, 2023.

[10] J. Xu *et al.*, "Qwen3-omni technical report," *arXiv preprint arXiv:2509.17765*, 2025.

[11] Y. Chu *et al.*, "Qwen2-audio technical report," *arXiv preprint arXiv:2407.10759*, 2024.

[12] Z. Cheng *et al.*, "Emotion-LLaMA: Multimodal emotion recognition and reasoning with instruction tuning," in *Adv. Neural Inf. Process. Syst.*, vol. 37, 2024, pp. 110 805–110 853.

[13] Z. Lian *et al.*, "AffectGPT: Dataset and framework for explainable multimodal emotion recognition," *arXiv preprint arXiv:2407.07653*, 2024.

[14] A. Jacovi and Y. Goldberg, "Towards faithfully interpretable NLP systems: How should we define and evaluate faithfulness?" in *Proc. 58th Annu. Meeting Assoc. Comput. Linguistics*, 2020, pp. 4198–4205.

[15] M. Turpin *et al.*, "Language models don't always say what they think: Unfaithful explanations in chain-of-thought prompting," in *Adv. Neural Inf. Process. Syst.*, vol. 36, 2024, pp. 74 952–74 965.

[16] J. Tian *et al.*, "Can LLMs express their uncertainty? An empirical evaluation of confidence elicitation in LLMs," in *Proc. 12th Int. Conf. Learn. Represent. (ICLR)*, Vienna, Austria, 2024.

[17] A. T. Kalai *et al.*, "Why language models hallucinate," *arXiv preprint arXiv:2509.04664*, 2025. [Online]. Available: https://arxiv.org/abs/2509.04664

[18] A. Zadeh *et al.*, "MOSI: Multimodal corpus of sentiment intensity and subjectivity analysis in online opinion videos,"*arXiv preprint arXiv:1606.06259*, 2016.

[19] A. B. Zadeh *et al.*, "Multimodal language analysis in the wild: CMU-MOSEI dataset and interpretable dynamic fusion graph," in *Proc. 56th Annu. Meeting Assoc. Comput. Linguistics (ACL)*, Melbourne, Australia, 2018, pp. 2236–2246.

[20] W. Yu *et al.*, "CH-SIMS: A Chinese multimodal sentiment analysis dataset with fine-grained annotation of modality," in *Proc. 58th Annu. Meeting Assoc. Comput. Linguistics*, Online, 2020, pp. 3718–3727.

[21] Z. Han *et al.*, “Benchmarking and bridging emotion conflicts for multimodal emotion reasoning,” in *Proc. 33rd ACM Int. Conf. Multimedia (MM)*, Dublin, Ireland, 2025, pp. 5528–5537, doi: 10.1145/3746027.3754856.
[22] J. A. Russell, “A circumplex model of affect,” *Journal of Personality and Social Psychology*, vol. 39, no. 6, pp. 1161–1178, 1980.
[23] *Evaluation of Measurement Data—Guide to the Expression of Uncertainty in Measurement*, JCGM 100:2008, Joint Committee for Guides in Metrology (BIPM, IEC, IFCC, ISO, IUPAC, IUPAP, OIML), Sèvres, France, 2008.
[24] Y. Wang and M. Wu, “Evaluation of data inconsistency for multimodal sentiment analysis,” in *Proc. 2024 IEEE Int. Conf. Data Mining Workshops (ICDMW)*, Pisa, Italy, 2024, pp. 381–390.
[25] Z. Chen *et al.*, “Text-driven noise filtering and affective conflict mitigation network for multimodal sentiment analysis,” in *Proc. Int. Conf. Netw. Netw. Appl. (NaNA)*, 2025, pp. 372–378.
[26] J. Wang and Y. Jiang, “Multimodal sentiment analysis based on text augmentation with adaptive gating and hierarchical fusion,” in *Proc. 10th Int. Conf. Inf. Sci. Comput. Technol. Transp. (ISCTT)*, 2025, pp. 126–131.
[27] Y. Luo *et al.*, “TriagedMSA: Triaging sentimental disagreement in multimodal sentiment analysis,” *IEEE Trans. Affect. Comput.*, vol. 16, no. 3, pp. 1557–1569, July–Sept. 2025, doi: 10.1109/TAFFC.2024.3524789.
[28] Y. Gao, H. Wu, and L. Zhang, “Multi-level conflict-aware network for multi-modal sentiment analysis,” in *ICASSP 2025-2025 IEEE Int. Conf. Acoust., Speech Signal Process.*, Apr. 2025, pp. 1–5.
[29] L. Kuhn, Y. Gal, and S. Farquhar, “Semantic uncertainty: Linguistic invariances for uncertainty estimation in natural language generation,” in *Proc. 11th Int. Conf. Learn. Represent. (ICLR)*, Kigali, Rwanda, May 2023.
[30] X. Wang *et al.*, “Self-consistency improves chain of thought reasoning in language models,” in *Proc. 11th Int. Conf. Learn. Represent. (ICLR)*, Kigali, Rwanda, May 2023.
[31] K. Tian *et al.*, “Just ask for calibration: Strategies for eliciting calibrated confidence scores from language models fine-tuned with human feedback,” in *Proc. 2023 Conf. Empir. Methods Natural Lang. Process. (EMNLP)*, Singapore, Dec. 2023, pp. 5433–5442.
[32] M. Xiong *et al.*, “Can LLMs express their uncertainty? An empirical evaluation of confidence elicitation in LLMs,” in *Proc. 12th Int. Conf. Learn. Represent. (ICLR)*, Vienna, Austria, May 2024.
[33] Y. A. Yadkori *et al.*, “To believe or not to believe your LLM: Iterative prompting for estimating epistemic uncertainty,” *arXiv preprint arXiv:2406.02543*, 2024.
[34] T. Tomov, D. Fuchsgruber, T. Wollschläger, and S. Günnemann, “The illusion of certainty: Uncertainty quantification for LLMs fails under ambiguity,” *arXiv preprint arXiv:2511.04418*, 2024.
[35] H. Orgad *et al.*, “LLMs know more than they show: On the intrinsic representation of LLM hallucinations,” in *Int. Conf. Learn. Represent. (ICLR)*, Singapore, 2025.
[36] M. Gao *et al.*, “Insights into LLM long-context failures: When transformers know but don’t tell,” in *Findings Assoc. Comput. Linguistics: EMNLP 2024*, Miami, FL, USA, 2024, pp. 7611–7625.
[37] N. Belrose *et al.*, “Eliciting latent predictions from transformers with the tuned lens,” *arXiv preprint arXiv:2303.08112*, 2023.
[38] A. Ghandeharioun *et al.*, “Patchscopes: A unifying framework for inspecting hidden representations of language models,” in *Proc. 41st Int. Conf. Mach. Learn. (ICML)*, Vienna, Austria, 2024, pp. 15 508–15 530.
[39] H. Rashkin *et al.*, “Towards empathetic open-domain conversation models: A new benchmark and dataset,” in *Proc. 57th Annu. Meeting Assoc. Comput. Linguistics*, Florence, Italy, 2019, pp. 5370–5381.
[40] Z. Lin *et al.*, “MoEL: Mixture of empathetic listeners,” in *Proc. 2019 Conf. Empir. Methods Natural Lang. Process. 9th Int. Joint Conf. Natural Lang. Process. (EMNLP-IJCNLP)*, Hong Kong, China, 2019, pp. 121–132.
[41] Y. Qian, W.-N. Zhang, and T. Liu, “Harnessing the power of large language models for empathetic response generation: Empirical investigations and improvements,” in *Findings Assoc. Comput. Linguistics: EMNLP 2023*, Singapore, 2023, pp. 1170–1183.
[42] Y. K. Lee *et al.*, “Chain of empathy: Enhancing empathetic response of large language models based on psychotherapy models,” *arXiv preprint arXiv:2311.04915*, 2023.
[43] M. De Choudhury, S. R. Pendse, and N. Kumar, “Benefits and harms of large language models in digital mental health,” *arXiv preprint arXiv:2311.14693*, 2023.
[44] E. C. Stade *et al.*, “Large language models could change the future of behavioral healthcare: A proposal for responsible development and evaluation,” *NPJ Mental Health Res.*, vol. 3, no. 1, p. 12, 2024.
[45] S. Bai *et al.*, “Qwen3-VL technical report,” *arXiv preprint arXiv:2511.21631*, 2025.
[46] W. Wang *et al.*, “Internvl3.5: Advancing open-source multimodal models in versatility, reasoning, and efficiency,” *arXiv preprint arXiv:2508.18265*, 2025.
[47] J. Xin *et al.*, “The art of abstention: Selective prediction and error regularization for natural language processing,” in *Proc. 59th Annu. Meeting Assoc. Comput. Linguistics 11th Int. Joint Conf. Natural Lang. Process. (ACL-IJCNLP)*, 2021, pp. 1040–1051.
[48] S. Whitehead *et al.*, “Reliable visual question answering: Abstain rather than answer incorrectly,” in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Tel Aviv, Israel, 2022, pp. 146–162.
[49] V. Satopaa, J. Albrecht, D. Irwin, and B. Raghavan, “Finding a “kneedle” in a haystack: Detecting knee points in system behavior,” in *2011 31st International Conference on Distributed Computing Systems Workshops*, 2011, pp. 166–171.